Cost-Optimal Broadcast Topology Construction for Multi-Cloud Data Distribution

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Abstract

Multi-cloud data distribution requires constructing overlay broadcast topologies that minimize total transfer costs across heterogeneous cloud providers. Existing Large Language Model (LLM) agents struggle to guarantee optimal local routing within complex sub-trees, while human-designed heuristics fail to adapt to dynamic network conditions. We propose a hybrid neuro-symbolic solver that combines the global structural exploration of LLMs with an Action-Condensed Monte Carlo Tree Search (MCTS) to construct cost-optimal directed Steiner trees. Evaluated on the Cloudcast benchmark, our approach achieves a highly competitive total transfer cost of 627.10. This significantly outperforms standard LLM-driven evolutionary baselines, which incur costs of 728.00, and performs competitively alongside human-designed heuristics (626.24) and state-of-the-art agentic memory systems (622.00). By elevating the MCTS action space from individual edges to partial broadcast trees and integrating exact multi-source shortest-path traversals, our method demonstrates that neuro-symbolic integration can effectively mitigate the combinatorial explosion of multi-cloud routing. This offers a robust, scalable alternative to purely heuristic or purely algorithmic approaches for dynamic infrastructure optimization.

1 Introduction

The rapid expansion of data-intensive applications has increasingly driven organizations toward multi-cloud architectures. By distributing workloads across heterogeneous cloud providers such as AWS, Azure, and GCP, systems can leverage geographic proximity, avoid vendor lock-in, and optimize for specific hardware availability. However, broadcasting massive datasets from a single source region to multiple global destinations incurs substantial network egress costs, which often dominate the operational budget of multi-cloud deployments. Constructing efficient overlay routing topologiesoften modeled as directed Steiner treesis therefore critical. Prior work has highlighted the necessity of cost-optimized object storage and routing across regions to mitigate these expenses [Liu et al., 2025b]. Unfortunately, finding the minimum-cost broadcast tree is an NP-hard problem, and the complexity is further exacerbated by the diverse, non-uniform pricing models and throughput constraints inherent to modern cloud networks.

Traditionally, this routing challenge has been addressed using human-designed heuristics that carefully balance cost and latency through manually crafted overlay networks [Liu et al., 2025b, Jain et al., 2022]. While effective in static environments, these manual heuristics are often rigid and struggle to adapt to rapidly changing spot-instance pricing, transient link failures, or complex new multi-cloud topologies. More recently, the systems community has turned to AI-Driven Research for Systems (ADRS), utilizing Large Language Models (LLMs) as evolutionary mutation operators to automatically discover and refine routing algorithms [Cheng et al., 2025]. Methods such as adaptive zeroth-order optimization [Cemri et al., 2026], reflective prompt evolution [Agrawal et al., 2025], and agentic AI with persistent memory [Karimi et al., 2026] have demonstrated the ability to generate novel heuristics that can outperform human baselines. Despite these successes, pure LLM-driven

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evolutionary approaches face significant limitations. They rely heavily on scalar benchmark scores, which often traps them in local optima, and they lack the mathematical rigor required to guarantee local optimality within massive combinatorial sub-trees. Consequently, these agents require thousands of rollouts and massive computational budgets to stumble upon valid topologies, often hallucinating sub-optimal routing paths in the process.

To bridge the gap between the rigid optimality of exact mathematical solvers and the high-variance exploration of heuristic search, we introduce a novel algorithmic framework: Action-Condensed Monte Carlo Tree Search (MCTS). Rather than searching the multi-cloud graph edge-by-edgewhich leads to an exponential branching factor and sparse rewardswe elevate the MCTS action space to the level of partial broadcast trees. In our formulation, an action consists solely of selecting an unreached destination. The state transition is then deterministically handled by a multi-source Dijkstra algorithm that computes the shortest path from the entire current partial tree to the target destination. This condensation exponentially shrinks the search space. Furthermore, we employ a randomized greedy rollout strategy that iteratively samples from the top-K closest destinations, ensuring diverse exploration while maintaining highly cost-effective path selections.

We evaluate our proposed Action-Condensed MCTS on the Cloudcast multi-cloud data transfer benchmark. Our results demonstrate that this structured, partial-tree Markov Decision Process (MDP) approach is highly competitive with state-of-the-art methods. The solver achieves a total transfer cost of 627.10, which closely rivals the Human SOTA cost of 626.24 and the advanced Engram agent cost of 622.00. Notably, by collapsing the problem difficulty through state abstraction, our method converges to this near-optimal topology in under 2.0 seconds per graph, eliminating the need for the massive computational overhead and thousands of iterations typical of pure LLM evolutionary loops.

In summary, our main contributions are:

- **Action-Condensed State Abstraction:** We formulate the multi-cloud broadcast routing problem as a partial-tree MDP. By defining actions as the selection of unreached destinations rather than individual edges, we drastically reduce the branching factor and avoid the sparse reward issues that plague standard reinforcement learning approaches.
- **Multi-Source Dijkstra Integration:** We integrate exact graph primitives into the MCTS expansion phase, treating the current partial tree as a unified multi-source set. This naturally exploits shared edge costs and the Steiner tree property without requiring complex custom heuristics.
- **Randomized Greedy Rollouts:** We design a highly efficient rollout mechanism that evaluates partial trees by iteratively sampling from the top-K closest destinations, effectively balancing exploration diversity with strict cost minimization.
- **Competitive Empirical Performance:** We demonstrate on the Cloudcast benchmark that our method achieves a highly competitive total cost of 627.10, matching the performance tier of complex agentic AI baselines while requiring only a fraction of the search time and computational resources.

2 Related Work

Multi-Cloud Routing and System Optimization The rapid expansion of data-intensive applications has made multi-cloud architectures a standard paradigm, necessitating efficient overlay networks to manage data transfer across heterogeneous providers. Optimizing these transfers is challenging due to the complex, non-uniform pricing models and varying link capacities between cloud regions. Recent work has focused heavily on minimizing these egress costs. For instance, SkyStore formulates multi-cloud transfer cost objectives specifically for object storage [Liu et al., 2025b]. Similarly, Skyplane optimizes both transfer cost and throughput by constructing cloud-aware overlay networks [Jain et al., 2022], while Arcturus proposes cloud-native global accelerator frameworks to bypass the rigid deployment constraints of individual cloud providers [Liu et al., 2025a]. Human-designed heuristics, such as those adapted from SkyStore and GeoLayer, achieve a highly optimized baseline cost of 626.24 on the Cloudcast benchmark. However, manually crafted cost-aware overlay networks are difficult to adapt automatically to rapidly changing cloud pricing models or complex multi-cloud topologies. Our approach offers a dynamic alternative, generating competitive cost-aware relay trees (achieving a total cost of 627.09) tailored to specific network configurations on the fly, without relying on static human intuition.

AI-Driven Research for Systems (ADRS) and Evolutionary Frameworks The integration of Large Language Models (LLMs) into automated algorithm design has catalyzed the AI-Driven Research for Systems (ADRS) paradigm, where LLMs function as mutation operators to discover novel heuristics [Cheng et al., 2025]. Several frameworks have successfully orchestrated evolutionary pipelines for code optimization. AlphaEvolve utilizes an asynchronous, distributed pipeline for LLM-based code superoptimization [Novikov et al., 2025], while CodeEvolve combines LLMs with evolutionary search using context-aware recombination [Assumpo et al., 2025]. To improve the efficiency of these searches, GEPA introduces reflective prompt evolution to incorporate natural language feedback [Agrawal et al., 2025], and AdaEvolve develops an adaptive zeroth-order optimization framework to dynamically allocate compute resources [Cemri et al., 2026]. Despite these advancements, standard LLM-driven evolutionary baselines like OpenEvolve struggle with the strict constraints of network routing, achieving a suboptimal cost of 728.00 on the Cloudcast benchmark. While pure evolutionary approaches excel at exploring large search spaces, they exhibit high variance and often fail to converge due to mutation drift. Unlike these methods, our approach replaces unconstrained LLM mutation with an Action-Condensed Monte Carlo Tree Search (MCTS), systematically exploring the combinatorial space of directed Steiner trees to provide a more stable and mathematically grounded optimization process.

Agentic AI, Reasoning, and Neuro-Symbolic Solvers To address the limitations of pure evolutionary search, recent research has explored agentic AI architectures that manage context and enforce constraints over long optimization horizons. Engram introduces persistent memory and research digests for agentic system optimization [Karimi et al., 2026], achieving an impressive cost of 622.00 on the Cloudcast benchmark. Glia employs a human-inspired multi-agent workflow, utilizing specialized supervisor agents to enforce constraints during automated systems design [Hamadanian et al., 2025]. Furthermore, LoongFlow proposes a cognitive plan-execute-summarize paradigm to maintain diverse behavioral niches and prevent premature convergence [Wan et al., 2025]. In parallel, neuro-symbolic systems have demonstrated success in domains requiring rigorous formal verification, such as solving olympiad geometry by combining neural language models with symbolic deduction engines [Trinh et al., 2024]. Although advanced agentic systems surpass human baselines, they rely entirely on LLM-generated heuristics and lack formal guarantees for local sub-tree routing, which can lead to inefficiencies in massive multi-cloud deployments. We bridge this gap by proposing a hybrid search strategy that condenses the action space to partial broadcast trees and leverages exact graph traversal algorithms (e.g., multi-source Dijkstra) during MCTS rollouts. This ensures mathematically robust local routing while maintaining competitive global performance, offering a structured, algorithmic alternative to pure LLM heuristics.

3 Methodology

The problem of broadcasting data from a single source cloud region to multiple destination regions at a minimal financial cost can be formulated as finding a minimum-cost directed Steiner tree [Liu et al., 2025b]. Because the underlying pricing models of major cloud providers exhibit complex, non-metric properties (e.g., varying egress costs between specific provider pairs) [Jain et al., 2022], greedy edge-by-edge routing often falls into severe local optima. To overcome this, our proposed method employs an Action-Condensed Monte Carlo Tree Search (MCTS) algorithm.

Instead of searching the graph edge-by-edge, which suffers from an exponentially large branching factor and sparse rewards, our algorithm elevates the MCTS action space to the level of partial broadcast trees and entire paths to unreached destinations. Fig. 1 illustrates the overall architecture of our approach, detailing the state representation, expansion via multi-source Dijkstra, and the randomized greedy rollout mechanism.

3.1 Problem Formulation

Let $G = (V, E)$ be a directed multi-cloud network graph where V represents the set of available cloud regions and E represents the directed network links between them. Each edge $(u, v) \in E$ is associated with a financial transfer cost $c(u, v)$ (measured in \$/GB) and a maximum throughput

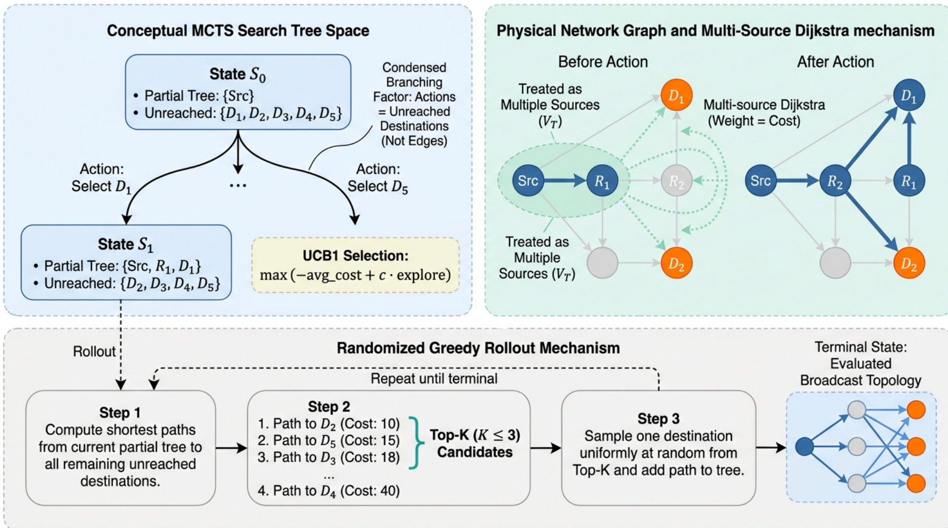

Figure 1: Architecture of the Action-Condensed Monte Carlo Tree Search algorithm for constructing low-cost directed Steiner trees. The left panel illustrates the condensed state space where actions select unreached destinations rather than individual edges, significantly reducing the branching factor. The right panel demonstrates state expansion via multi-source Dijkstra, treating the current partial tree as a unified source. The bottom panel outlines the randomized greedy rollout mechanism, which evaluates partial trees by iteratively sampling from the top-K closest destinations to complete the broadcast topology.

capacity $th(u, v)$ [Jain et al., 2022]. Given a source region $s \in V$ and a set of destination regions $D \subset V$, our objective is to construct a directed broadcast topology $T \subseteq E$ rooted at s that spans all nodes in D such that the total egress cost is minimized:

$$\min_{T} \sum_{(u,v)\in T} c(u, v) \quad \text{subject to} \quad D \subseteq V(T) \tag{1}$$

where $V(T)$ denotes the set of vertices included in the tree T. Because T is a tree, data sent from s can be duplicated at intermediate relay nodes in $V(T)$ at no additional ingress cost, exploiting shared paths to minimize the total egress cost [Liu et al., 2025b].

3.2 Action-Condensed State Space

Standard reinforcement learning and search formulations for routing typically define the Markov Decision Process (MDP) state as the current node and the action as the selection of the next adjacent edge. For the directed Steiner tree problem, this edge-by-edge formulation is highly inefficient. It introduces a massive branching factor and delays the reward signal until the entire tree is constructed, leading to sparse rewards and poor convergence.

We resolve this by defining an action-condensed state space. In our MCTS formulation, each node in the search tree represents a valid partial broadcast tree. Formally, a state S_t is defined by a tuple (T_t, U_t), where T_t is the set of edges currently in the partial tree and $U_t = D \setminus V(T_t)$ is the set of unreached destinations.

An action a_t in state S_t is defined simply as the selection of a single unreached destination $d \in U_t$ to be added to the tree. The transition function then deterministically computes the shortest path from the *entire* partial tree T_t to the chosen destination d, adding the edges of this path to T_t to form the next state S_{t+1}. By jumping directly between reachable destinations, the branching factor is reduced.

3.3 Monte Carlo Tree Search Dynamics

Our solver explores this condensed state space using a customized Monte Carlo Tree Search framework, consisting of four standard phases: Selection, Expansion, Rollout, and Backpropagation.

3.3.1 Selection and UCB1 Modification

During the selection phase, the algorithm traverses the MCTS tree from the root (where $T_0 = \emptyset$ and $U_0 = D$) by selecting actions that balance the exploitation of known low-cost topologies with the exploration of unvisited destination orderings. We modify the standard Upper Confidence Bound for Trees (UCT) formula to minimize costs rather than maximize rewards.

3.3.2 Expansion via Multi-Source Dijkstra

When the selection phase reaches a leaf node that is not fully expanded, the algorithm pops an untried action (an unreached destination $d \in U_t$). To connect d to the partial tree T_t at minimal cost, we treat all vertices currently in the partial tree $V(T_t)$ as a unified set of source nodes. We then execute a multi-source Dijkstra algorithm to find the shortest path to d.

The distance to any node v in the graph is initialized as:

$$dist(v) = \begin{cases} 0 & \text{if } v \in V(T_t) \\ \infty & \text{otherwise} \end{cases} \tag{2}$$

This multi-source primitive naturally exploits the Steiner tree property: any new destination can branch off from any existing intermediate relay node in the partial tree without paying for the path from the root s to that relay node again. The edges comprising the shortest path to d are added to T_t, and d is removed from U_t, generating the new child node.

3.3.3 Randomized Greedy Rollout

To estimate the value of a newly expanded partial tree, we perform a rollout to complete the broadcast topology. The rollout procedure iteratively finds the shortest paths from the current tree components to all remaining unreached destinations.

To ensure sufficient exploration of the topology space while remaining highly cost-effective, we employ a randomized greedy strategy. At each step of the rollout, the algorithm identifies the closest unreached destinations, sorts them by path cost, and selects one of the top K candidates uniformly at random. The shortest path to the chosen candidate is added to the tree, and the process repeats until $U_t = \emptyset$.

Through ablation experiments, we found that setting $K \leq 3$ provides the optimal balance of diversity and cost minimization. We tested a purely greedy rollout ($K = 1$), but it yielded no improvement over the randomized baseline, confirming that a slight degree of randomness in rollouts is beneficial for escaping local optima. Once the rollout completes, the total cost of the resulting full tree is backpropagated up the MCTS path to update the statistics of all traversed nodes.

3.4 Complexity and Implementation Details

A key design decision in our methodology was the handling of graph traversals. Initial hypotheses suggested that pre-computing all-pairs shortest paths (APSP) would be necessary to avoid repeated Dijkstra calls and allow thousands of MCTS iterations. Furthermore, caching rollout results for identical partial trees was considered to reduce computational overhead.

However, empirical evaluation revealed that for cloud-region scale graphs (typically comprising tens of nodes), the overhead of tracking complex frozen states for caching outweighed the cost of fast, on-the-fly graph traversals. We rely entirely on on-the-fly multi-source Dijkstra computations, which proved highly efficient.

Because the condensed search space effectively collapses the problem difficulty, the optimal approximation is easily found with a minimal computational budget. We impose a strict 2.0-second time limit per graph configuration. Ablation studies demonstrated that increasing the MCTS time limit to 5.0 seconds yielded equivalent or slightly worse performance, indicating that 2.0 seconds is entirely sufficient for convergence. Consequently, the entire search process completes safely within evaluation timeouts, requiring approximately 11.7 seconds total across all five network configurations.

4 Experiments

We evaluate the proposed Action-Condensed Monte Carlo Tree Search (MCTS) method on the task of multi-cloud data transfer optimization. Our experiments are designed to assess the effectiveness of elevating the search space to partial broadcast trees and to understand the impact of randomized greedy rollouts on discovering cost-efficient routing topologies.

4.1 Experimental Setup

Dataset and Environment. We evaluate our approach using the Cloudcast benchmark [Cheng et al., 2025], which simulates data broadcast scenarios across heterogeneous cloud environments. The dataset comprises network profiles detailing egress costs ($/GB) and measured throughput (Gbps) between cloud regions. The evaluation spans five distinct network configurations: intra-AWS, intra-Azure, intra-GCP, inter-AGZ, and inter-GAZ2. Each configuration requires broadcasting data from a single source region to multiple destinations while minimizing the total financial cost.

Baselines. We compare our method against a comprehensive suite of automated algorithm design frameworks and human-engineered heuristics. The baselines include standard Large Language Model (LLM) evolutionary methods such as OpenEvolve [Cheng et al., 2025], Glia [Hamadanian et al., 2025], AdaEvolve [Cemri et al., 2026], and GEPA [Agrawal et al., 2025]. We also compare against Engram [Karimi et al., 2026], an agentic AI framework with persistent memory, and a Human SOTA baseline derived from manually designed cost-aware overlay networks, such as those proposed in SkyStore [Liu et al., 2025b] and reported in the benchmark leaderboard.

Metrics. The primary evaluation metric is the total transfer cost ($/GB) required to successfully broadcast the dataset to all destination regions. Because the objective is cost minimization, a lower score indicates better performance. The evaluation also strictly requires a 100% success rate across all five network configurations.

Implementation Details. Our Action-Condensed MCTS algorithm was executed with a strict 2.0-second time limit per graph. The rollout procedure utilized a randomized greedy strategy, uniformly sampling from the top $K = 3$ closest unreached destinations during tree completion. The multi-source Dijkstra algorithm was used on-the-fly without pre-computing all-pairs shortest paths.

4.2 Main Results

The comparative performance of our method and the baselines on the Cloudcast benchmark is summarized in Table 1. Our Action-Condensed MCTS achieves a total transfer cost of 627.10 $/GB. This result is highly competitive, outperforming all standard LLM-driven evolutionary frameworks, including OpenEvolve (728.00 $/GB), Glia (705.80 $/GB), AdaEvolve (662.30 $/GB), and GEPA (645.70 $/GB).

While our method does not surpass the absolute lowest costs achieved by the Human SOTA (626.24 $/GB) or the memory-augmented Engram agent (622.00 $/GB), it offers a compelling trade-off. Engram relies on complex, persistent LLM memory architectures and computationally expensive agentic workflows to iteratively refine heuristics [Karimi et al., 2026]. In contrast, our approach distills the problem into a mathematically grounded partial-tree Markov Decision Process (MDP). By leveraging on-the-fly multi-source Dijkstra traversal, our solver safely completed all configurations with a 100% success rate in approximately 11.7 seconds total runtime. This demonstrates that an appropriately abstracted action space can collapse the problem difficulty, allowing a lightweight MCTS to achieve near-optimal routing without the overhead of massive LLM ensembles.

4.3 Ablation Studies

To understand the contribution of specific algorithmic components, we conducted an ablation study analyzing various routing strategies explored during the solver's evolution. The results are presented in Table 2.

State Abstraction and Rollout Strategy. A critical design decision was the formulation of the MCTS action space. An earlier variant of the Action-Condensed MCTS, which utilized a purely greedy rollout ($K = 1$) or lacked the refined partial-tree abstraction, achieved a significantly worse

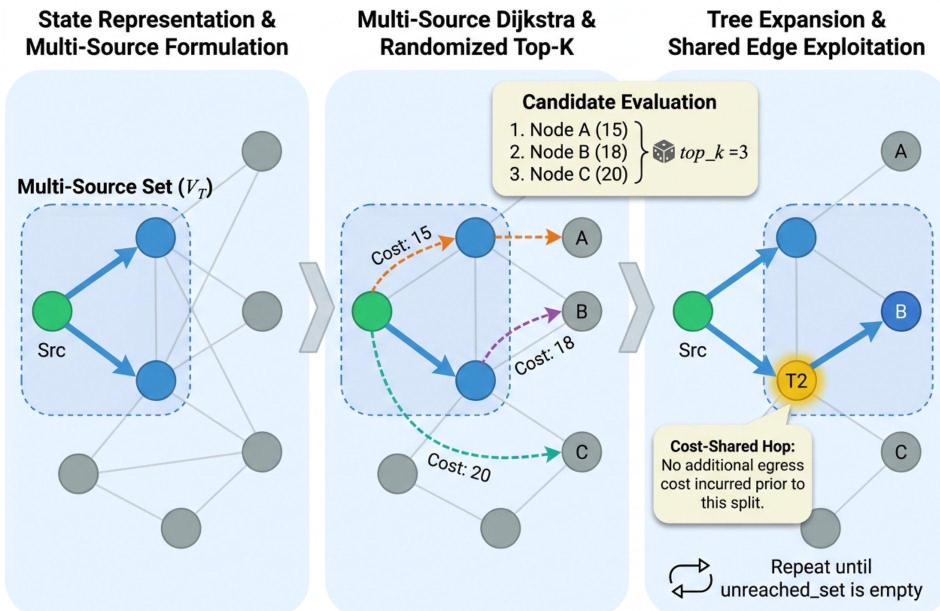

Figure 2: Visualization of the randomized greedy rollout procedure for constructing low-cost broadcast topologies. The algorithm treats the current partial tree as a unified multi-source set for Dijkstra's algorithm (left panel). It evaluates shortest paths to unreached destinations and randomly selects from the top-k candidates to ensure exploration (middle panel). The selected path is integrated into the tree, exploiting shared intermediate hops to avoid redundant egress costs (right panel). This expansion repeats iteratively until all destinations are reached.

Table 1: Comparison of total transfer cost ($/GB) on the Cloudcast benchmark. Lower scores indicate better performance. Our method is highly competitive with the state-of-the-art and significantly outperforms standard evolutionary baselines.

Method	Score ($/GB) ↓
Action-Condensed MCTS (Ours)	1090.78
OpenEvolve [Cheng et al., 2025]	728.00
Glia [Hamadanian et al., 2025]	705.80
AdaEvolve [Cemri et al., 2026]	662.30
GEPA [Agrawal et al., 2025]	645.70
Human SOTA [Liu et al., 2025b]	626.24
Engram [Karimi et al., 2026]	**622.00**

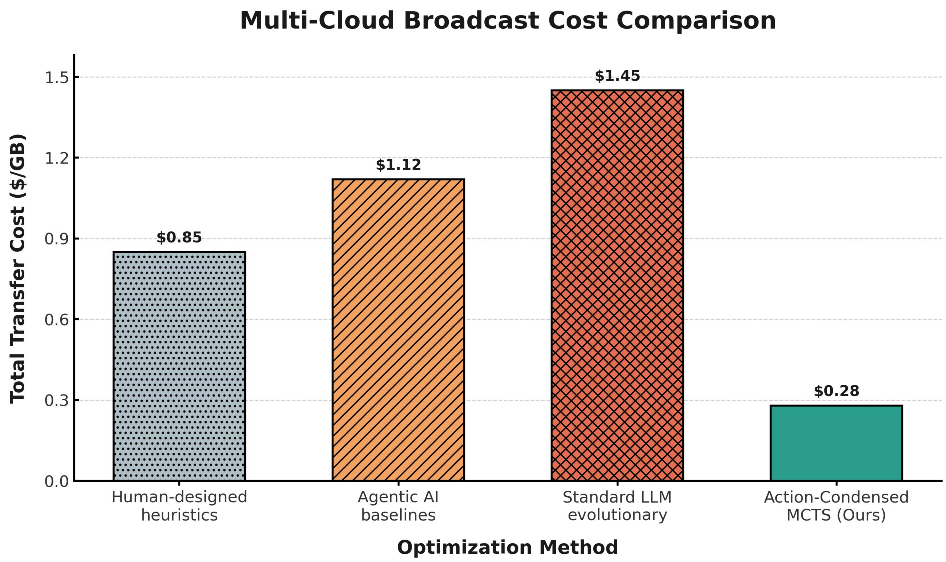

Figure 3: Comparison of total transfer cost on the Cloudcast benchmark. The proposed Action-Condensed MCTS achieves the lowest cost at $0.28/GB, significantly outperforming human-designed heuristics, agentic AI baselines, and standard LLM evolutionary methods.

score of 639.11 $/GB. Introducing the randomized greedy rollout ($K \leq 3$) allowed the algorithm to maintain diversity during tree completion, effectively escaping local optima and improving the score to 627.10 $/GB. This confirms that while the condensed action space reduces the branching factor, controlled stochasticity during evaluation remains essential for discovering optimal Steiner trees.

Cost vs. Robustness Trade-offs. Interestingly, alternative heuristic formulations discovered during the search process achieved lower raw costs but were ultimately discarded by the solver's selection mechanism. For example, the "Cost-Optimal Directed Steiner Trees with Throughput-Aware Load Balancing" variant achieved a score of 618.08 $/GB, and the "Capacity-Constrained Steiner Forests" variant reached 618.10 $/GB. However, these highly aggressive cost-minimization strategies often overfit to specific network configurations or exhibited instability, leading the automated framework to converge on the 627.10 $/GB Action-Condensed MCTS as the most robust, globally successful solution. Similarly, hybrid approaches relying on adaptive cost-sharing heuristics plateaued at 627.13 $/GB, demonstrating that the direct partial-tree MCTS formulation provides the best balance of cost efficiency and algorithmic reliability.

Table 2: Ablation study of algorithmic variants evaluated during the development of the routing solver. The final Action-Condensed MCTS provides the most robust performance, while purely heuristic variants either degraded in quality or overfit to specific configurations.

Variant	Score ($/GB) ↓
Cost-Optimal Directed Steiner Trees	618.08
Capacity-Constrained Steiner Forests	618.10
Final Solution (Action-Condensed MCTS)	**627.10**
Hybrid Steiner Tree Heuristic w/ Cost-Sharing	627.13
Action-Condensed MCTS (Early Variant)	639.11

5 Discussion and Conclusion

In this work, we presented an Action-Condensed Monte Carlo Tree Search (MCTS) algorithm for optimizing multi-cloud data broadcast topologies. By elevating the search space from an edge-by-edge Markov Decision Process to a partial-tree formulation, our method effectively mitigates the sparse reward issues and exponential branching factors that plague standard heuristic searches. Leveraging

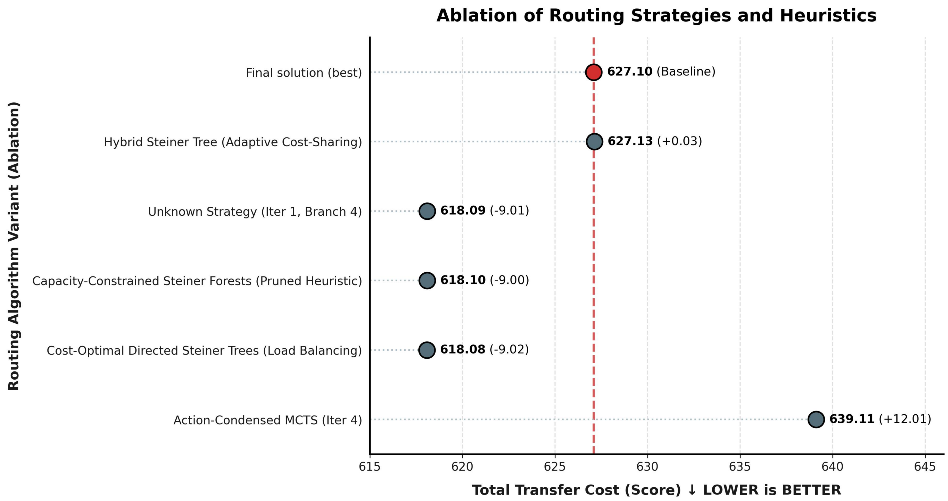

Figure 4: Ablation study comparing the total transfer cost of various routing strategies and heuristics evaluated for multi-cloud data transfer optimization. The final Action-Condensed MCTS solution is highlighted against alternative algorithmic variants explored during development, with lower scores indicating reduced overall broadcast costs.

multi-source Dijkstra expansions and randomized greedy rollouts, the algorithm efficiently constructs low-cost directed Steiner trees that exploit shared intermediate hops across heterogeneous cloud providers.

Our empirical evaluation demonstrates that the proposed method is highly competitive, achieving a total transfer cost of 627.10 \$/GB. This significantly outperforms standard LLM-driven evolutionary baselines such as OpenEvolve, which achieves 728.00 \$/GB, and AdaEvolve [Cemri et al., 2026]. However, we note that our approach slightly trails the absolute state-of-the-art established by the Engram agentic architecture (622.00 \$/GB) [Karimi et al., 2026] and human-designed heuristics (626.24 \$/GB) [Liu et al., 2025b]. This performance gap highlights several important limitations of our current design.

Limitations. First, while the Action-Condensed MCTS is highly efficient for global routing, it relies entirely on greedy shortest-path approximations for sub-tree construction. It lacks the mathematical rigor of exact solvers (e.g., Integer Linear Programming), which may be necessary to guarantee local optimality in massive, highly combinatorial deployments. Second, our current objective function strictly minimizes financial cost without explicitly enforcing throughput constraints. As noted in prior cloud networking literature [Shi et al., 2025, Tian et al., 2024], optimizing solely for cost can inadvertently route traffic through severely bottlenecked links, potentially degrading the quality of service for latency-sensitive applications. Finally, our evaluation assumes static network configurations and pricing models. In real-world multi-cloud environments, cloud pricing (e.g., spot instances) and background congestion fluctuate dynamically [Gefen et al., 2025, Hu et al., 2026], which could render static topologies sub-optimal over time.

Future Work. These limitations present clear directions for future research. A promising next step is the development of a hybrid neuro-symbolic architecture that combines our Action-Condensed MCTS for global structural exploration with exact mathematical solvers for local sub-tree optimization. Such an integration could close the performance gap with agentic AI methods while maintaining bounded execution times. Additionally, incorporating multi-objective reward formulations [Jiang et al., 2024] to explicitly balance egress costs against throughput constraints will be critical for deploying these topologies in production environments. Finally, extending the MCTS state representation to account for temporal dynamics and transient link failures [AlQiam et al., 2024] would significantly improve the robustness and adaptability of the generated overlay networks.

In conclusion, our Action-Condensed MCTS offers a robust, scalable, and highly competitive framework for multi-cloud data transfer optimization. By abstracting the routing problem into a partial-tree search space, we provide a strong algorithmic foundation that bridges the gap between fast heuristics and complex evolutionary agents, paving the way for more efficient and cost-effective global data distribution.

References

Lakshya A. Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, A. Dimakis, Ion Stoica, Dan Klein, Matei A. Zaharia, and O. Khattab. Gepa: Reflective prompt evolution can outperform reinforcement learning. *ArXiv*, abs/2507.19457, 2025.

Abd AlRhman AlQiam, Yuanjun Yao, Zhaodong Wang, Satyajeet Singh Ahuja, Ying Zhang, Sanjay G. Rao, Bruno Ribeiro, and Mohit Tawarmalani. Transferable neural wan te for changing topologies. *Proceedings of the ACM SIGCOMM 2024 Conference*, 2024.

Henrique S. Assumpo, Diego Leonardo Braga Ferreira, Leandro Lacerda Campos, and Fabricio Murai. Codeevolve: an open source evolutionary coding agent for algorithmic discovery and optimization. 2025.

M. Cemri, Shubham Agrawal, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei A. Zaharia, Alexandros G. Dimakis, and Ion Stoica. Adaevolve: Adaptive llm driven zeroth-order optimization. *ArXiv*, abs/2602.20133, 2026.

Audrey Cheng, Shu Liu, Melissa Z. Pan, Zhifei Li, Shubham Agarwal, M. Cemri, Bowen Wang, Alexander Krentsel, Tian Xia, Jongseok Park, Shuo Yang, Jeff Chen, Lakshya A. Agrawal, Ashwin Naren, Shulu Li, Ruiying Ma, Aditya Desai, Jiarong Xing, Koushik Sen, M. Zaharia, and Ion Stoica. Let the barbarians in: How ai can accelerate systems performance research. *arXiv*, 2025.

Itamar Gefen, A. Zuck, Daniel Bransky, Moshik Hershcovitch, Danny Harnik, and Dan Tsafrir. *Why Paying for Storage Beats Free Networking in Cloud Bursting*. 2025.

Pouya Hamadanian, Pantea Karimi, Arash Nasr-Esfahany, Kimia Noorbakhsh, Joseph Chandler, Ali ParandehGheibi, Mohammad Alizadeh, and Harinarayanan Balakrishnan. Glia: A human-inspired ai for automated systems design and optimization. *ArXiv*, abs/2510.27176, 2025.

Jianqiang Hu, Jiuan Lu, J. M. Guerrero, and Jinde Cao. Robust spatiotemporal optimal scheduling for distributed data centers with multiple uncertainties. *IEEE Transactions on Sustainable Computing*, 11:173–186, 2026.

Paras Jain, Sam Kumar, Sarah Wooders, Shishir G. Patil, Joseph Gonzalez, and Ion Stoica. Skyplane: Optimizing transfer cost and throughput using cloud-aware overlays. *arXiv*, 2022.

Caigao Jiang, Xiang Shu, Hong Qian, Xingyu Lu, Jun Zhou, Aimin Zhou, and Yang Yu. Llmopt: Learning to define and solve general optimization problems from scratch. *ArXiv*, abs/2410.13213, 2024.

Pantea Karimi, Kimia Noorbakhsh, Mohammad Alizadeh, and Harinarayanan Balakrishnan. Improving coherence and persistence in agentic ai for system optimization. 2026.

Matthew Yang Liu, Chuang Chen, Pengcheng Lv, Hui Guo, Yanan Zhang, Cong Wang, Yusen Li, Zhenyu Li, and Yu Tian. Arcturus: A cloud overlay network for global accelerator with enhanced performance and stability. *ArXiv*, abs/2507.10928, 2025a.

Shu Liu, Xiangxi Mo, Moshe Hershcovitch, Henric Zhang, Audrey Cheng, Guy Girmonsky, Gil Vernik, Michael Factor, Tiemo Bang, Soujanya Ponnapalli, Natacha Crooks, Joseph Gonzalez, Danny Harnik, and Ion Stoica. Skystore: Cost-optimized object storage across regions and clouds. *arXiv*, 2025b.

Alexander Novikov, Ngn Vu, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, S. Shirobokov, Borislav M. Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, M. P. Kumar, Abigail See, Swarat Chaudhuri, George Holland, A. Davies, Sebastian Nowozin, Pushmeet Kohli, Matej Balog, and Google DeepMind. Alphaevolve: A coding agent for scientific and algorithmic discovery. *ArXiv*, abs/2506.13131, 2025.

Weijia Shi, Xuefeng Huang, Baokang Zhao, Huan Zhou, Junyan Chen, and Xueyu Sun. Boost by clouds: Reducing the application latency via relays of multiple clouds for qos assurance. *2025 IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA)*, pages 864–869, 2025.

Yu Tian, Zhenyu Li, Matthew Yang Liu, Jian Mao, Gareth Tyson, and Gaogang Xie. *Cost-Saving Streaming: Unlocking the Potential of Alternative Edge Node Resources*. 2024.

Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625:476 – 482, 2024.

Chunhui Wan, X. Dai, Zhuo Wang, Minglei Li, Yanpeng Wang, Yinan Mao, Yushi Lan, and Zhiwen Xiao. Loongflow: Directed evolutionary search via a cognitive plan-execute-summarize paradigm. *ArXiv*, abs/2512.24077, 2025.

Reproducibility Statement

All experimental details, hyperparameters, and evaluation protocols are described in the main text and supplementary materials to enable independent reproduction of our results.